① Qiskit program

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)          fault
7 circuit.cx(0,1)
8 circuit.measure_all()
```

lines 10–16: simulate, print

Expected $(|01\rangle+|10\rangle)/\sqrt{2}$*; line 6 should apply X.*

② srcML of line 7

```
<expr_stmt pos:start="7:1"[d]>
 <expr><call>[a]
  <name><name>circuit</name>
   <operator>.</operator><name>cx[b]</name></name>
  <argument_list>(<argument><expr>
   <literal type="number">0[c]</literal>
   </expr></argument>,<argument><expr>
   <literal type="number">1</literal>
   </expr></argument>)</argument_list>
 </call></expr></expr_stmt>
```

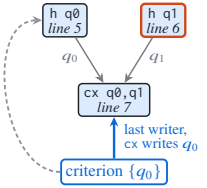
a `call` → one operation **b** `cx` → gate
c `0`, `1` → q_0 control, q_1 target
d position → line 7 the QDG writes both

③ Trace

line	gate	qubits
5	h	q_0
6	h	q_1
7	cx	q_0, q_1
8	measure	q_0
8	measure	q_1

Read statically. Grey measurements are terminal.

④ QDG



Shaded nodes are the QDG's backward reach, and the dashed arrow is the classical PDG's last writer.

⑤ Slice of q_0

QDG PDG

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
7 circuit.cx(0,1)
8 circuit.measure_all()
```

QDG keeps the faulty line 6
classical PDG keeps line 5 only

A full square marks a quantum line, a light one a classical line that builds the circuit.